Full-path and hypergraph Christofides comparison

Created-at: 2026-10-22 02:05 CEST

ETA: approximately 60–90 minutes: implementation and correctness checks, frozen Linux build, guarded five-process controls and reporting. Host contention can delay measurement.

Owner and scope Graph owns the additive Christofides algorithm over canonical Core IR. Core and Tensor remain pinned dependencies. Existing Graph evidence clients, fixture builders, external adapters and guarded campaign runner own this experiment. No additional IR, scheduling engine, dependency, commit or publication.

Accepted problem User confirmed a closed walk visiting every vertex, with repeated original vertices and hyperedges permitted. Costs are finite, nonnegative and symmetric, one traversal cost per hyperarc. Christofides operates on shortest Berge distances; the expanded walk retains incidence identities. The classical metric 3/2 bound applies to this walk objective, not a simple Berge cycle. Exact minimum matching is initially bounded subset dynamic programming; exceeding its explicit state budget returns an error, never greedy matching under the Christofides name.

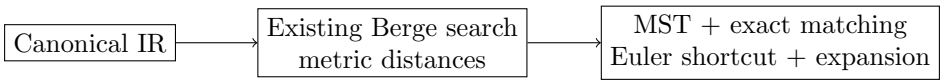
Implementation Add a cohesive bounded Christofides owner in direct_ir, reusing the existing Dijkstra engine for metric closure and expansion. Extend the informed evidence client with first-result Dijkstra, full path exports and one small genuine multiway transport tour. Extend the existing external adapter with library-owned memberships plus explicitly labelled NetworkX weighted routing/Christofides composition. Record runtime, hops, total cost, route validity and tour coverage. Separate available native capabilities from composed workflows.

Files direct_ir/mod.rs and tour.rs; focused tour integration tests; informed_berge_evidence.rs and its assisted/routing/reporting modules; scripts/performance_evidence/campaign.py, external.py, routing.py and their tests; README.md and CORE.YAML. Reports and frozen input exports under reports/2026-10-22-path-tour-comparison.

Approval and base User explicitly requested the new case and measurements. CORE.YAML approval records this scope. Graph feat/newest-core at ea1c588; fetch-all succeeded this phase, upstream equal, origin/main tree identical. Preserve existing dirty files. Public API is additive.

Verification Small exact TSP oracle, independent NetworkX Christofides, all-node coverage, closed walk, every incidence and cost, disconnected/empty/singleton/zero-cost/invalid-weight cases and budget rejection. Source and canonical-input hashes, no cached answers. Normal tests never measure. Pilot each new path before full measurements. Five processes, two warmups, nine samples; CPU 2, nice 10, 180 seconds/process and 16 GiB cap; retain quiet-host guard.

Risk and rollback Metric closure is quadratic and the first matching backend exponential in odd-degree vertices; report limits explicitly and reject before oversized allocations. Floating-point costs retain documented arithmetic tolerance; dyadic fixture weights allow exact independent comparisons. Foreign workflows may share NetworkX and are labelled as composition, not independent native TSP implementations. Rollback only the named task edits, preserving all previous work. Archive one consolidated PDF/data/plots phase after writers finish.



Scaling continuation – 2026-10-22, 03:30 CEST

User explicitly requests larger graph and TSP measurements overnight, followed by releasing the lab machine and closing our connections. ETA: approximately 60–90 minutes including preparation/reporting; the remote build and measurement process group has a hard 60-minute cap. Reuse the same Graph checkout and evidence owners. Fetch-all succeeded again; branch/upstream equal and origin/main tree identical. Production solver and dependency pins stay unchanged.

Inputs: a 512×512 grid (262144 nodes, 511-hop row query), weighted 256×256 roads, 256×256 transport with off-station endpoints; 24-, 32-, 64-node street/transport tours and a 64-node corridor/transport structural control. All originate in existing canonical-IR fixtures. The original 16-node tour remains a correctness anchor. The 64-node street case is retained even if exact subset matching reaches its default 2^{20}-state limit.

Extend existing Scenario/Model fixture metadata and neutral export, Campaign selection, external route verification and tests. Use explicit heuristic metadata so cheap transport never acquires an invalid geometric lower bound. For tours beyond the 16-node exact-oracle scope, independently derive MST and maximum one-tree lower bounds from the metric; report cost/lower-bound as a certified upper bound on the optimum ratio, never as an exact optimality gap. Preserve every original incidence in returned witnesses. Resource-budget rejection is a labelled result, not a fallback algorithm.

Pilot candidate strategies first, freeze the fastest successful candidate per workload/framework for five independent measured processes, and retain all pilot/limit results. Skip failed pilot cells in the full campaign. Use the existing Campaign, ProcessRunner, QuietHostGuard and ControlSession; CPU 2, nice 10, 16 GiB, 180 seconds/process, 15 seconds/query for external clients. Do not weaken the quiet gate. Enforce the global deadline on our process group, collect results, verify no owned workload remains, and use nonpersistent SSH sessions. Never stop another user's processes or the user's own interactive login.

Correctness covers new shapes/endpoints, admissible heuristics, matching-budget classification, full route checks and lower-bound validation. Reports, plots and closure receipts go under reports/2026-10-22-path-tour-scaling and are archived once. No commit, push, merge or public upload is authorized by this continuation.

Budget adaptation, 04:01 CEST Larger external process costs justify prioritizing TSP and the 65536-node controls. For the 262144-node input, complete five-process controls compare HyMeKo with the fastest external pilot candidate (Reticula native unit-distance route). Other strategies on that largest input remain explicitly pilot-only. Preserve the original one-hour process-group deadline. Pause only the waiting parent controller while its pilot continues; after a complete pilot, resume the same frozen sources and binary with reordered jobs and the remaining original time budget.

Closing allowance, 04:41 CEST The user's overnight authorization permits completion of the requested controls. Quiet-host waits consumed the initial allowance. Announce and bound one final extension to 65 minutes total elapsed remote time, without changing CPU count, priority, memory or the quiet gate. Let the original supervisor stop at its deadline, preserve any interrupted log, then resume only incomplete cells from the validated checkpoint. No further extension is planned; close all owned workloads/connections after collection.